

Single Source Justification (Simplified Acquisition)**Section A: General Contract Information**Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)?☐ Yes☒ No

Contracting Activity: AFLCMC/PZIBA

Project/Program Name: Adelphi High Output DD Neutron Generator

Purchase Request / Local ID Number: [REDACTED]

Estimated Contract Cost (including Options): [REDACTED]

Section B: Description of the Supplies/Services Required

This requirement was initiated by Air Force Institute of Technology (AFIT) at Wright-Patterson AFB for the acquisition of a High Output Neutron Generator. Specifically, a neutron generator that causes fusion between deuterium atoms, emitting 1E10 neutrons per second is required.

Section C: Justification for Soliciting from a Single Source or Brand Name

The output of this generator is unmatched by any other companies. The high output of this generator will reduce measurement and experiment times by at least 100x compared to neutron generators available from other companies. Additionally, Adelphi is able to support a maintenance contract to ensure this generator operates at maximum efficiency. This generator will directly support students in at least three nuclear courses: NENG612, NENG650, and NENG812. The energies of the neutrons emitted by a DD generator are well-defined and unique to that interaction, thus various types of research and studies can be performed. Additionally, a generator with this capacity will be of interest to other organizations conducting neutron research, providing an avenue for funding and collaboration. The high output will be especially beneficial for these relationships.

Section D: Efforts to Obtain Competition

Market research was performed and only Adelphi is able to offer a neutron generator with this capability. Other companies researched include [REDACTED]. Both offer a DD neutron generator, but with a maximum output of 1E8 neutrons per second. Other companies, such as [REDACTED], offer DD neutron generators with outputs comparable to the intensity desired, but for other reasons were not considered. [REDACTED] is a company located [REDACTED] mentions on their website that they are able to support the maintenance of equipment in the [REDACTED]. Having foreign nationals come to perform maintenance in our research area is not feasible. [REDACTED] was not considered as their contact information and website appeared out of date. Additionally [REDACTED] not mention that they offer maintenance of their equipment, which is a necessity to ensure it is fully functional over time. A Sources Sought notice was posted to SAM.gov and only Adelphi provided a response.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

The Government will continue to monitor the state-of-the-art systems being developed through website searches, reading published articles, and attendance at conferences and trade shows.

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Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(2\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

